



NIT Kurukshetra

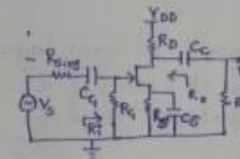
End Sem Exam Analog Electronics (ECPC-40) IV Sem.

Duration: 2 Hr

MH: 50

Q.1. Discuss in detail factors that define the -3dB Cut-off
(a) Point of CE amplifier at high frequency. (6)

Q.1 (b) Determine the lower Cutoff frequency for the given network using the following Parameters: (4)



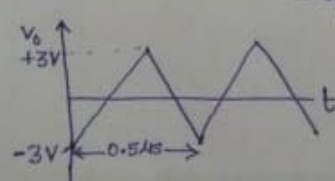
$C_E = 0.01 \mu F$; $C_C = 0.5 \mu F$; $C_S = 2 \mu F$
 $R_{S_{eq}} = 10 k\Omega$; $R_B = 1 M\Omega$; $R_D = 4.7 k\Omega$; $R_E = 1 k\Omega$; $R_L = 22 k\Omega$
 $I_{DSS} = 8 mA$; $V_P = -4 V$; $r_d = \infty \Omega$; $V_{DD} = 20 V$.

Q.2. Describe Wilson Current Mirror Circuit (4)

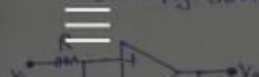
Q.2 (a) What are the various Configuration of differential amplifier
(b) also find value of voltage gain A_v and input impedance R_i for any one configuration using AC analysis. (6)

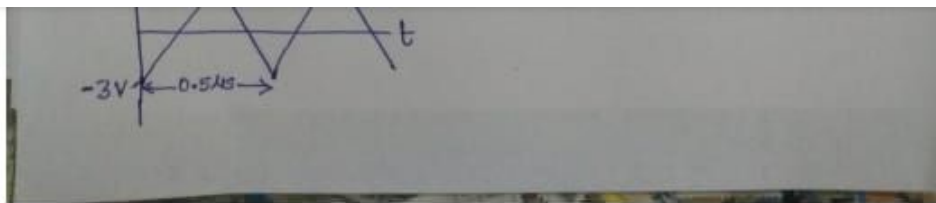
Q.3. Derive expression for voltage gain and input resistance for
(a) differential amplifier with two op-amp. (5)

Q.3 (b) What is slew rate and its importance? The OP of an op-amp voltage follower is a triangular wave as shown in fig. for a square wave i/p of frequency 2 MHz and 8V peak to peak amplitude. What is the slew rate of the op-amp? (3+2)

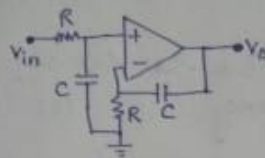



Q.4 (a) In the given fig. derive the expression of V_O . (5)





Q.4 (a) In the given fig. derive the expression of V_o . (5)



Q.4 (b) Design a differentiator to differentiate an i/p signal that varies in frequency from 20 Hz to about 2 kHz. If a square wave of 2V peak to peak at 2 kHz is applied to i/p of designed ckt. draw its o/p wave form. (5)

Q.5 (a) Design a second order low pass filter at a high cutoff frequency of 10 kHz. (4)

Q.5 (b) Derive the expression for frequency of oscillation f_o and condition for stability ($R_f = 2R_1$) for Wien bridge oscillator. (6)